

Confidential Containers

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Agenda

1. What is a Container ?
2. What is Kubernetes?
3. What is a Virtual Machine?
4. Why do we need extra security?
5. What is Confidential Computing?
6. What are Confidential Containers?

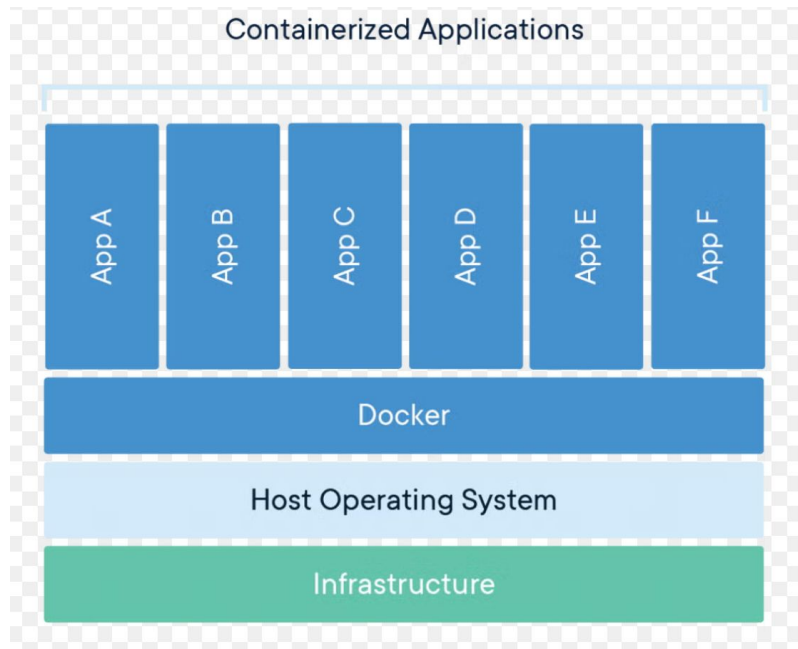


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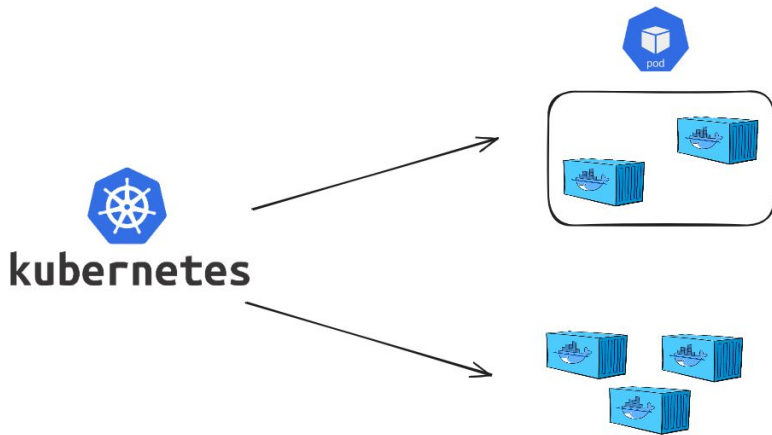
What Is a Container?

- A container is a package for running an application.
- It contains the application and its required files.
- It helps the application run the same way on different systems.
- Containers are lightweight and fast to start.
- Multiple containers can run on one server.
- Containers share the host operating system kernel.



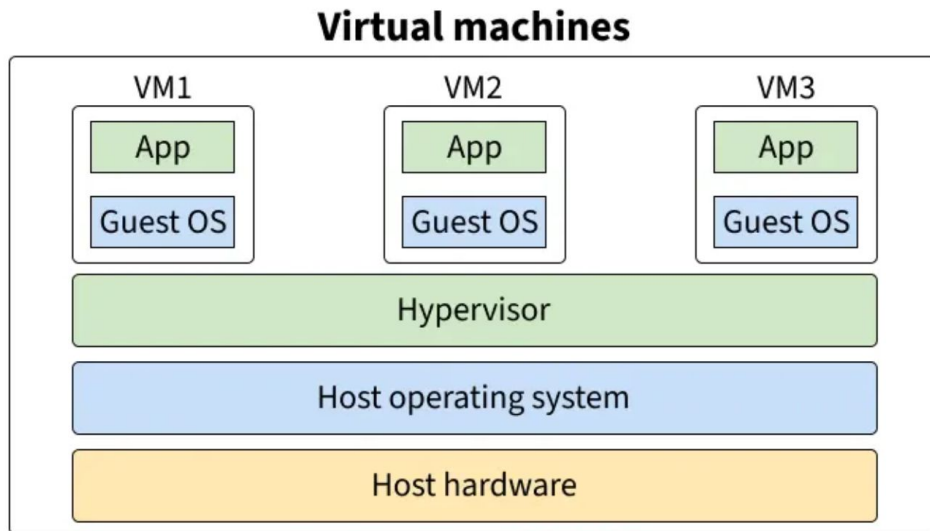
What Is Kubernetes?

- Kubernetes is a platform for running container applications.
- It helps deploy and manage many containers.
- It can start, stop, and restart applications.
- It can run applications on multiple servers.
- In Kubernetes, an application usually runs inside a Pod.
- A Pod can contain one or more containers



What Is a Virtual Machine?

- A Virtual Machine, or VM, is like a separate computer inside a server.
- A VM has its own operating system.
- A VM has its own memory and file system.
- VMs are managed by a hypervisor.
- A VM gives stronger separation from the host.
- A VM uses more resources than a container.



Why Do We Need Extra Security?

- Some applications use sensitive data.
- Examples: passwords, encryption keys, customer data, and AI models.
- Normal containers provide application isolation.
- VMs provide stronger isolation.
- But sensitive data may need protection while it is running.
- This is where Confidential Computing helps.



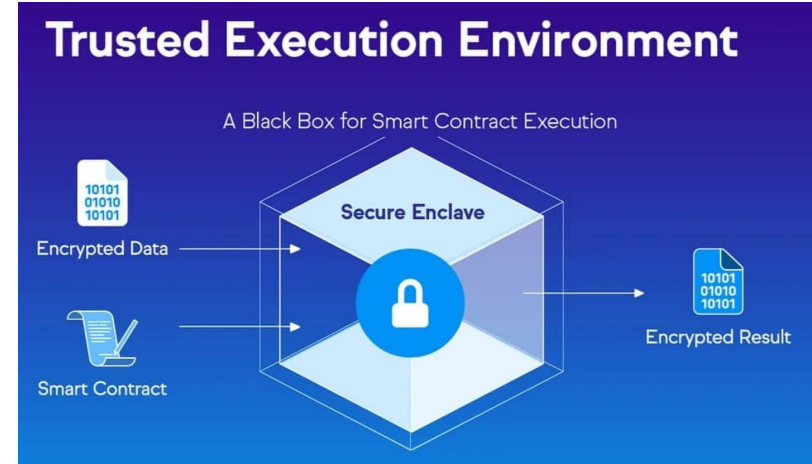
What Is Confidential Computing?

- Confidential Computing protects data while it is being used.
- This is also called protecting data in use.
- It uses security features built into the processor.
- It creates a protected area for the workload.
- The host system should not read the protected workload memory.
- This protected area is called a TEE.



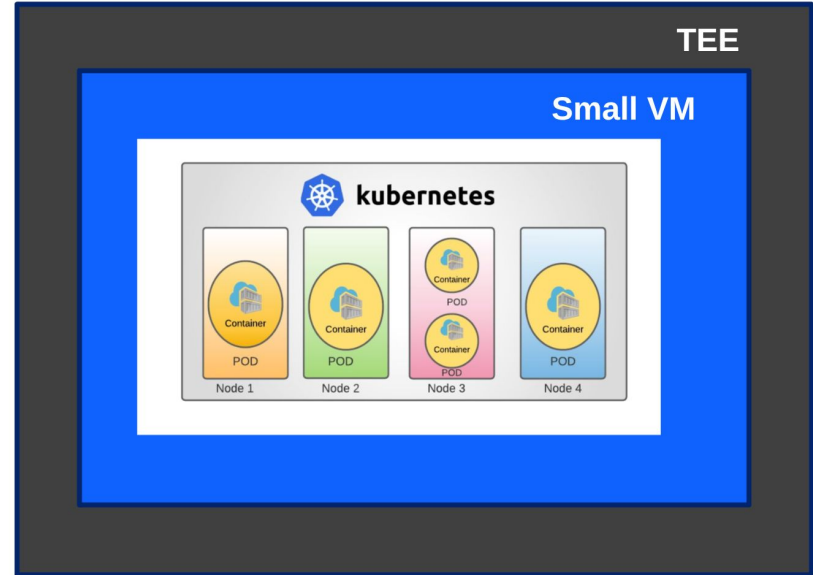
What Is a TEE?

- TEE means Trusted Execution Environment.
- It is a protected area created by hardware.
- Applications can run inside this protected area.
- The hardware protects the workload memory.
- The host OS and hypervisor are outside the TEE.
- A TEE adds an extra security boundary.



What Are Confidential Containers?

- Confidential Containers are also called CoCo.
- CoCo brings Confidential Computing to Kubernetes.
- A Kubernetes Pod is run inside a small VM.
- The VM is protected using a Trusted Execution Environment (TEE).
- The application runs inside this protected VM.
- This gives extra protection to the application and its data

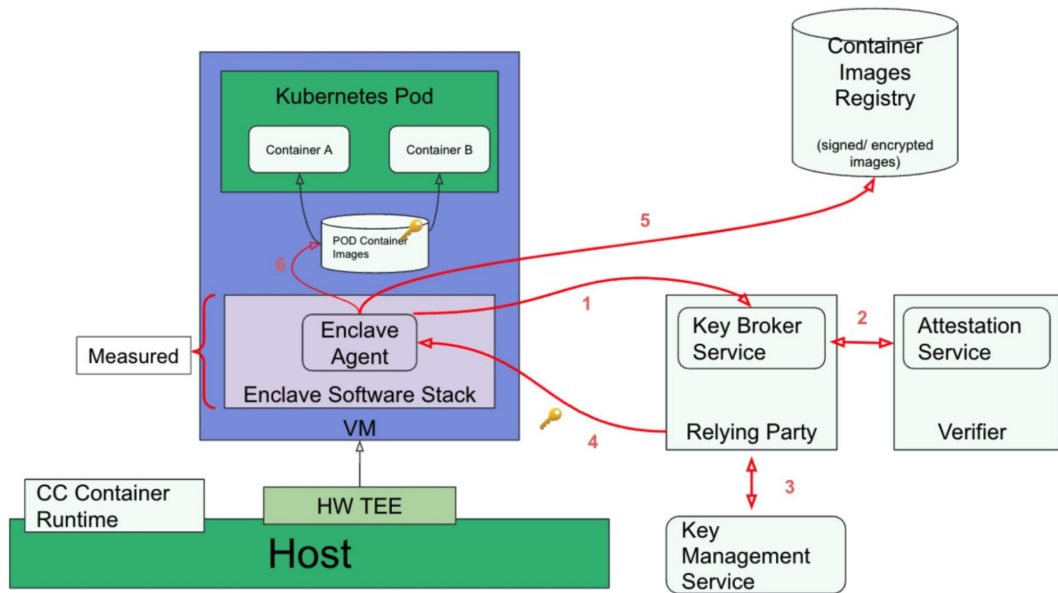


What Is Attestation?

- Attestation is a security verification check.
- The protected VM sends security proof.
- The proof shows that trusted hardware is being used.
- It also shows that expected software is running.
- A trusted service checks this proof.
- Secrets are shared only when the check passes.

How Does CoCo Run a Pod?

- Kubernetes receives a request to run a Pod.
- Kubernetes selects a node for the Pod.
- The CoCo runtime creates a small VM for that Pod.
- The container runs inside the small VM.
- The VM runs inside the TEE.
- Kata Containers is commonly used as the VM-based runtime.



Key Takeaway

- Kubernetes manages container applications.
- Containers package and run applications.
- VMs provide stronger isolation.
- Confidential Computing protects data in use.
- CoCo runs a Kubernetes Pod inside a protected VM.
- Attestation checks the VM before secrets are released.

